

The Future of PRAYCG — Public GitHub Roadmap v1.2

Version: 1.2 public clean

Prepared: 2026-08-27T19:25:03Z

Status: Public-facing research roadmap for GitHub and OSF.

Executive summary

PRAYCG should advance as a staged open-methods program for naturalistic psychophysiology. The project is strongest when it is presented not as a finished theory of consciousness, but as a reproducible workbench for testing how sensory entrainment, narrative reception, analytic task stance, artifact, respiration, and after-state dynamics interact.

The next phase should move from single-operator exploratory runs toward controlled individual replication, then small group replication, then confound-expanded variants using higher-order stimulus controls, eye tracking, EOG/EMG, counterbalanced order, and improved timing validation.

Current safest thesis

PRAYCG is a theory of availability:
whether a living system is available to decode, receive, integrate, and be changed by a meaningful stimulus.

This thesis does not require proving microtubules, biophotons, quantum biology, metaphysics, or clinical mechanism. It requires showing that state trajectories differ across carefully controlled conditions.

What should remain stable

The core PRAYCG structure should remain:

- PhaseScrambled Control
- Target
- Contextual Override
- Baseline and washout windows
- Final reflection baseline
- Physical display-timing validation
- Artifact/confound reporting
- Self-report as independent context

Future work should not remove the phase-scrambled Control. It is not a perfect meaning-null branch, but it is the first defense against sensory-entrainment overinterpretation.

Why phase-scrambled Control remains necessary

A movie can drive EEG and autonomic changes through luminance, cuts, audio envelope, rhythm, faces, salience, and shared attention. The phase-scrambled branch asks whether the Target response is reducible to low-level audiovisual drive.

The correct interpretation is not:

Control is meaningless.

The correct interpretation is:

Control is meaning-damaged but sensory-active.

Why phase-scrambled Control is not sufficient

Phase scrambling also damages high-order visual structure: faces, bodies, edges, biological motion, and object continuity. Therefore Target > PhaseScrambled could reflect intact face/object/social perception rather than narrative order.

This motivates the high-order structural-control variant.

Variant 4: High-Order Visual Control / Structural Defense

Structure

PhaseScrambled -> ShotOrderScramble -> Target -> Override

Rationale

ShotOrderScramble preserves local shots, faces, bodies, objects, voice fragments, and biological motion while disrupting the narrative temporal arc. It is not a replacement for phase scrambling. It is a second control layer.

Stronger future contrast

Target > PhaseScrambled
Target > ShotOrderScramble
Target > Override

at locked anchors and after timing/artifact/confound gates.

Correct claim

If Target survives ShotOrderScramble, the result becomes less likely to be explained by faces, bodies, objects, or local biological motion alone, and more consistent with narrative-order or temporal-meaning structure.

Incorrect claim

It would still not “definitively prove meaning.” It would strengthen the interpretation while leaving open artifact, gaze, order, familiarity, and physiology confounds.

Four-day individual study design

Day 1 — Acclimation and setup

- Fit cap and sensors.
- Test ALS holder and runner-side barcode.
- Run LabRecorder and stream checklist.

- Run short non-emotional stimulus test.
- Confirm participant can tolerate setup.

Day 2 — Calibration battery

- Eyes-open baseline.
- Neutral breathing.
- Paced breathing.
- Mild cognitive load.
- Neutral naturalistic media.
- Artifact provocation: blink, jaw, eyes, motion labels.
- Optional EOG/EMG calibration.

Day 3 — Hypothesis-critical PRAYCG run

- Frame-locked anchors.
- PRAYCG2.1 runner-side barcode.
- PhaseScrambled, Target, Override.
- Optional ShotOrder branch if PRAYCG2.2 is available.
- Full self-report/confound reports.
- No public claim if timing/QC fails.

Day 4 — Recovery / retest

- Baseline retest.
- Short Target-memory/reactivation check.
- Debrief.
- Withdrawal option.
- Confound review.
- Post-run qualitative interview.

Group-study roadmap

Stage 0 — Software/hardware validation

Run synthetic and local pilot workflows until:

- ALS barcode passes,
- stream watchdog passes,
- MediaPrep outputs validate,
- Control audio QC passes,
- visual QC passes,

- offline interpreter outputs are understandable.

Stage 1 — Small N feasibility

Use 8–12 participants to estimate variance, artifact rate, task compliance, comfort, and timing success.

Stage 2 — Counterbalanced replication

Use counterbalanced orders:

```
Control -> Target -> Override
Control -> Override -> Target
PhaseScrambled -> ShotOrder -> Target -> Override
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Stage 3 — Confirmatory endpoint

Preregister:

- primary anchors,
- bands/channels/features,
- timing gates,
- artifact thresholds,
- Target > Control / Target > ShotOrder / Target > Override contrasts,
- self-report as covariate rather than proof.

Required hardware upgrades

High-priority:

- EOG
- EMG/jaw or forehead artifact channel
- robust ALS/PT19 holder
- runner-side ALS barcode
- stable OpenBCI stream monitoring
- eye tracking or at least webcam gaze proxy

Useful:

- EDA/GSR
- pupilometry
- higher-density EEG
- better headphones or controlled audio

Software roadmap

PRAYCG2.1

Runner-side ALS barcode before each branch.

PRAYCG2.2

Four-branch protocol support:

PhaseScrambled
ShotOrderScramble
Target
Override

MediaPrep v1.9+

ShotOrderScramble generator:

- detect shots,
- shuffle shot order,
- apply same cue schedule after shuffle,
- write shot table and QC report.

Master Suite v1.6+

Unify:

- A-MRED,
- MRED-Peak/Resolution,
- NUPI,
- DGA,
- HOC-R,
- OSA,
- OHC,
- AAM,
- RespDualPath,
- ALS barcode detector,
- ShotOrder condition handling.

Institutional deployment

The best academic home is not only psychology and not only computer science. The ideal home is:

cognitive neuroscience / neuroaesthetics / media psychology
+ biomedical engineering / signal processing / machine learning collaboration

The psychological side is needed for stimulus design, self-report, experience, narrative theory, and ethics. The engineering side is needed for timing, artifacts, LSL, signal processing, classifiers, and reproducibility.

Ethics and public release

- Do not use copyrighted stimuli in public repositories.
- Do not publish raw private biosignal data without consent and proper review.
- Do not call self-run data confirmatory.
- Do not present speculative mechanism as established biology.
- Keep OSM/cellular/microtubule content quarantined from PRAYCG empirical claims.
- Use clear participant consent and debriefing for any human study.

Bottom line

PRAYCG's future is not to become a grand proof machine. Its future is to become a stricter measurement workbench:

Can we tell when the nervous system is tracking sensory structure,
when it is receiving meaning,
when task stance changes reception,
and when an after-state persists beyond the stimulus?

That is enough. It is testable, useful, and worth building.